

Discrete Mathematics

Master of Computer Applications (MCA) — Semester 1 Complete Digital Study Notes

Course Code: MCA-101 / MCA-102

Format: Digital Notes & Examples

UNIT 1: SET THEORY, RELATIONS & MATHEMATICAL LOGIC

1. Sets & Relations

A **Set** is a collection of distinct, well-defined objects. A **Relation** R from set A to set B is a subset of the Cartesian product $A \times B$.

Properties of Binary Relations on a Set A :

- Reflexive** $\forall a \in A, (a, a) \in R$
- Symmetric** If $(a, b) \in R \implies (b, a) \in R$
- Transitive** If $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$
- Antisymmetric** If $(a, b) \in R$ and $(b, a) \in R \implies a = b$

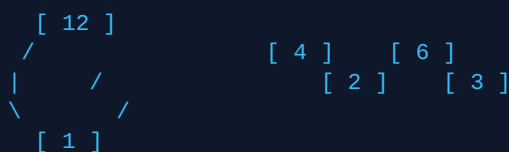
2. Posets & Hasse Diagrams

A relation R on set A is a **Partially Ordered Set (POSET)** if it is Reflexive, Antisymmetric, and Transitive. It is denoted as $(A, \leq)$.

EXAMPLE: HASSE DIAGRAM CONSTRUCTION

Problem: Draw the Hasse Diagram for the poset $(D_{12}, |)$, where $D_{12} = \{1, 2, 3, 4, 6, 12\}$ and relation is divisibility ($a | b$).

Solution / Diagram Structure:



Note: In a Hasse diagram, reflexive loops and transitive edges are omitted, and directions move strictly upward.

3. Propositional Logic & Truth Tables

A proposition is a declarative statement that is either True (T) or False (F).

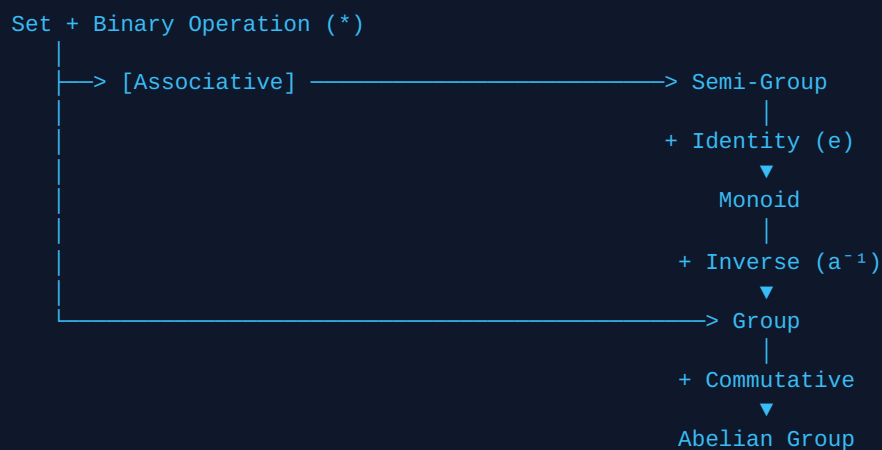
P	Q	$P \wedge Q$ (AND)	$P \vee Q$ (OR)	$P \rightarrow Q$ (Implication)	$P \leftrightarrow Q$ (Biconditional)
T	T	T	T	T	T

P	Q	$P \wedge Q$ (AND)	$P \vee Q$ (OR)	$P \rightarrow Q$ (Implication)	$P \leftrightarrow Q$ (Biconditional)
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

UNIT 2: ALGEBRAIC STRUCTURES, LATTICES & BOOLEAN ALGEBRA

1. Algebraic Structures Hierarchy

An algebraic structure consists of a non-empty set equipped with one or more binary operations satisfying specific axioms:



2. Lattices & Boolean Algebra

A lattice is a POSET $(L, \leq)$ in which every pair of elements $a, b \in L$ has a unique **Least Upper Bound (LUB / Join $\vee$)** and **Greatest Lower Bound (GLB / Meet $\wedge$)**.

EXAMPLE: K-MAP MINIMIZATION

Problem: Simplify the Boolean function $F(A, B) = A'B + AB + AB'$ using a 2-variable Karnaugh Map.

	B = 0	B = 1	
A = 0	0	1	<-- A'B
A = 1	1	1	<-- AB', AB

Grouping:

- Group 1 (Row $A = 1$): Covers $AB' + AB = A$
- Group 2 (Column $B = 1$): Covers $A'B + AB = B$

Minimal Expression: $F(A, B) = A + B$

UNIT 3: COMBINATORICS & RECURRENCE RELATIONS

1. Pigeonhole Principle

If $k + 1$ or more objects are placed into k boxes, then at least one box contains two or more objects.

EXAMPLE PROBLEM

Question: How many people must be in a room to guarantee that at least two share the same birth month?

Solution: Here, number of boxes (months) $k = 12$. By Pigeonhole Principle, we need $n = 12 + 1 = 13$ people.

2. Recurrence Relations

A recurrence relation defines a sequence a_n using previous terms.

SOLVED STEP-BY-STEP PROBLEM

Solve: $a_n - 5a_{n-1} + 6a_{n-2} = 0$ given initial conditions $a_0 = 1, a_1 = 4$.

Step 1: Characteristic Equation

Replace a_n with r^n :

$$r^2 - 5r + 6 = 0 \implies (r - 2)(r - 3) = 0$$

Roots: $r_1 = 2, r_2 = 3$ (Real and Distinct)

Step 2: General Solution Form

$$a_n = C_1(2)^n + C_2(3)^n$$

Step 3: Apply Boundary Conditions

For $n = 0$: $C_1 + C_2 = 1$

For $n = 1$: $2C_1 + 3C_2 = 4$

Solving equations gives $C_1 = -1$ and $C_2 = 2$.

Final Solution:

$$a_n = -(2^n) + 2(3^n)$$

UNIT 4: GRAPH THEORY & TREES

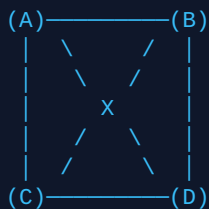
1. Fundamentals & Handshaking Lemma

A graph $G = (V, E)$ consists of vertices V and edges E .

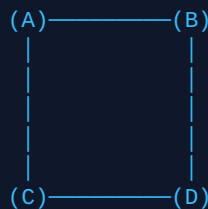
Handshaking Lemma: For any undirected graph G :

$$\sum \deg(v) = 2 |E| \text{ (Sum of degrees of all vertices equals twice the number of edges)}$$

2. Graphs Visualizations



Complete Graph K_4 (6 Edges)



Planar Representation (No Edge Crossings)

3. Euler vs. Hamiltonian Graphs

- **Euler Path / Circuit:** Traverses every *edge* of the graph exactly once. (Exists if every vertex has an even degree).
- **Hamiltonian Path / Circuit:** Visits every *vertex* of the graph exactly once.